

CableTwin

"See before you decide" — the Waze of the cable plant

Phase 1 — The Idea

AI & Automation Hackathon "AUTOMATE OR DIE" 2026 — Sector 3: Industry · Theme 3: Production planning & supply chain

Concept note (Phase 1) · July 17, 2026 · Track 1 — One-Person Company

"When a line stops, CableTwin turns a production emergency into three comparable decisions. The production manager sees the threatened orders, the delays, the costs and the changes before validating a new plan."

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Number integrity: every figure in this document comes from a synthetic, reproducible simulated scenario (Appendix A). No figure is presented as performance measured in a real factory.

Executive summary

In a cable factory, an unplanned line stop triggers a negotiation under pressure: phone calls, spreadsheets, the planner's memory. The production manager must decide fast, without an overall view, and the decision is neither compared against alternatives nor traced. **CableTwin is a decision twin**: the moment an incident occurs, it shows the threatened orders and the impact of doing nothing, then generates **three comparable recovery plans** — computed under real constraints and evaluated on several objectives (delay, cost, disruption). The human compares, arbitrates and validates; every decision is logged.

CableTwin's claim in Phase 1 is a **decision outcome**, not factory ROI. The value unit: after an unplanned line stop, produce three constraint-feasible recovery plans, compare them on delay, overtime, transfers and extra cost, then record a human-approved choice. On synthetic scenario S1 only — 3 lines, 10 orders of which 3 are replannable, 11 feasible plans scored exhaustively by the runnable V0 engine — inaction projects **620 minutes** of delay (7/10 on time); the Speed profile reaches **140 minutes** (9/10); the Balance trade-off reaches **150 minutes** at a lower extra cost *than inaction*. Those figures prove the engine can surface trade-offs a planner cannot rebuild by phone and spreadsheet under pressure. They are not measured plant savings, not customer ROI, and not industrial validation: real impact is defined only by the pilot's go/no-go thresholds, measured on the partner's own historical incidents (\$4.3, \$9).

1 incident → 3 plans

constraint-feasible, compared, human-approved, 100% logged — the unit of value

620 → 140 min

S1 engine proof — synthetic: 3 replannable orders, 11 plans scored

Impact: pilot

industrial impact is measured only in the pilot — shadow replay vs recorded decisions (\$4.3, \$9)

1. The problem

A cable factory runs firm-deadline orders across a few expensive lines (extrusion, cabling, armouring). When a line stops — machine failure, material defect, a key operator missing — the schedule becomes wrong at that very instant, and three urgent questions arise: *which orders are threatened? what happens if we do nothing? which reorganization should we choose?*

Today the answer is handcrafted:

- **Slow** — rebuilding a plan by hand (spreadsheet, phone, shop-floor meeting) typically takes one to several hours (*business assumption to be validated in the pilot, \$4*), during which the healthy lines sometimes produce the wrong priorities.
- **Blind** — a single option is built, never compared: nobody knows what it costs, what an alternative would have yielded, or even the true cost of inaction.
- **Invisible to the customer** — delays are discovered order by order, too late to warn or renegotiate.
- **Untraced** — impossible to audit why an order was sacrificed, or to learn from one incident to the next.

The person who suffers this problem is the **production manager** (together with the planner and the shop-floor supervisor): they arbitrate under stress, commit overtime and own the customer delay. The cable and wire-harness sector — a flagship export industry in Tunisia and beyond — lives these stops daily, mostly without an APS or any decision-support tool.

Anchoring in the official theme. Sector 3 — Industry · Theme 3 "Production planning and supply chain", axis "Production scheduling and capacity optimization". The targeted business process is **production replanning after an unplanned line stop**. CableTwin does not predict the failure (theme 1): it optimizes the response when it happens — complementary approaches, with repair-duration prediction on our roadmap. Answers to the theme's questions: *the bottlenecks?* the stopped line and the remaining lines' capacities/compatibilities; *demand data?* the dated order book, imported read-only (CSV/ERP) — no forecasting required in V1; *impact on lead times and costs?* measured on S1: 620 → 140/150 minutes of delay and extra costs compared plan by plan (\$4).

2. The solution

CableTwin is the Waze of the cable plant. When an obstacle appears, Waze does not merely report the traffic jam: it recomputes routes, states each one's cost in minutes and lets the driver choose. CableTwin does exactly that for a workshop: an operational model of the plant (lines, compatibilities, orders, deadlines) which, at incident time, produces comparable recovery plans — and lets the human decide. **"See before you decide."**

Product journey (the very one that will be demonstrated end-to-end in Phase 2):

initial schedule → **L2 stops** → impact with no action (threatened orders, projected delay) → **three strategies** generated under constraints → KPI comparison (delay, cost, transfers, overtime) → selection → **human validation** → time-stamped audit log → reset.

- **AI proposes, the human disposes:** no action is applied without explicit validation; each plan is a readable list of actions (transfers, resequencing, overtime).
- **Preview before deciding:** the revised schedule (Gantt) and the moved orders are shown *before* any approval — nothing changes until the human validates.
- **Audit log:** incident, alternatives shown, choice, author and timestamp are kept — the decision becomes an asset, no longer a reflex.
- **Offline, deterministic, resettable:** same inputs, same plans; both the demo and the future pilot can be replayed identically.
- **Read-only by design:** CableTwin writes nothing into the ERP/MES; it consumes exports (CSV, then connectors) — the condition for fast, risk-free adoption.

3. The innovation — where the AI really is

CableTwin's AI is neither a chatbot bolted onto a dashboard nor a black box. It is a **constraint-based plan-search engine coupled with multi-objective evaluation** — at the core of the product, not at its periphery:

1. **Generation under hard constraints** — the engine explores order reassignments across *compatible* lines (product families, per-line speeds), resequencing and bounded overtime, while respecting line availability and repair duration. An infeasible plan is never shown: constraints are guaranteed *by construction*.
2. **Multi-objective evaluation** — every feasible plan is scored on several axes: total delay, late orders, extra cost (overtime, transfers), disruption (number of line changes).
3. **Three contrasted profiles** — rather than a single, debatable "optimum", the engine returns three strategies embodying different trade-offs (Speed / Balance / Zero transfer), because the right choice depends on context only the human knows (strategic customer, crew fatigue, quality).

4. **Full explainability** — each strategy is deterministic, replayable, and justified by its KPIs and action list: the preconditions for trust and auditability in an industrial setting.

Proof that search beats intuition: when building scenario S1 by hand, our best "balanced" plan reached 170 minutes of delay. The engine found 150 minutes — by resequencing two orders in a counter-intuitive way (serving the latest-deadline order first). That is precisely the human blind spot the product exists to remove.

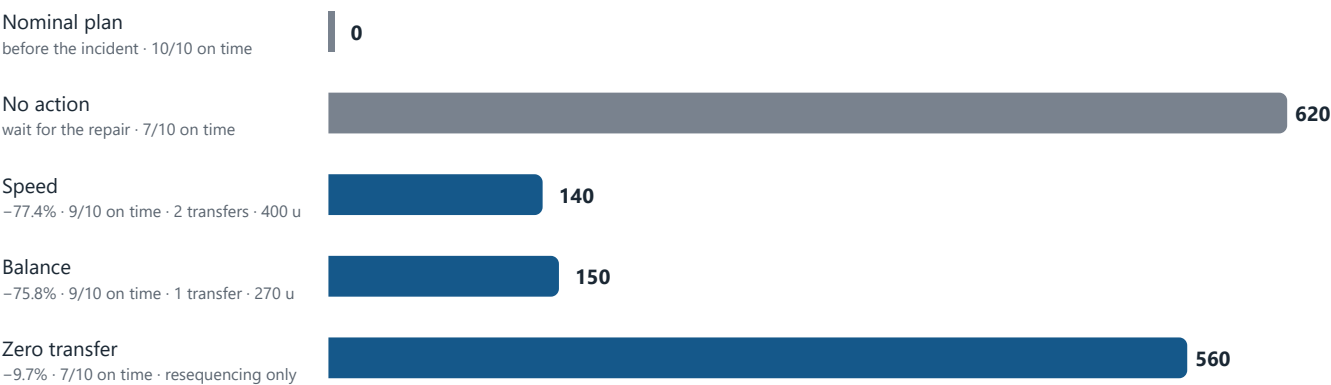
This AI belongs to a historic, recognized branch of artificial intelligence: **automated planning and constraint-based optimization** (symbolic AI, the family of constraint satisfaction and optimization solvers) — the same algorithmic family that powers routing engines like Waze, industrial solvers such as CP-SAT, and chess engines. V0, already coded, is an **exact solver**: a complete search of the plan space under hard constraints, with feasibility guaranteed *by construction* and per-objective optimality at scenario scale — where a probabilistic black-box approach would guarantee neither. At industrial scale (dozens of orders, an exponential combinatorial space), the same architecture shifts to a **multi-objective meta-heuristic search** (local search, simulated annealing), changing neither the constraint model nor the deterministic explainability production managers require; later, learning components (repair-time prediction, learning the decision-maker's preferences) will enrich the engine. *Generate under constraints, score multi-objective, let the human arbitrate*: the architecture stays identical at every scale.

4. Quantified impact

Number integrity. The results below are computed by the V0 engine (`sim/cabletwin_sim.py` , deterministic, standard Python, reproducible in one command) on the synthetic scenario S1 described in Appendix A. They are **simulation results**; the **business assumptions** and the **pilot target KPIs** are presented separately. No figure in this document is a performance measured in a real factory.

4.1 Simulation results (scenario S1 — 11 feasible plans evaluated)

Projected total delay per plan (minutes) — synthetic scenario S1



V0 engine output on S1 (`sim/results.json`). Extra costs in synthetic units (u): line transfer 150 u, overtime 2 u/min. Simulation results — not real-world measurements.

Plan	Total delay	On time (of 10)	Transfers	Overtime	Synthetic extra cost
Nominal plan — before the incident	0 min	10 / 10	0	0 min	0 u
No action (reference)	620 min	7 / 10	0	150 min	300 u
Speed — minimal delay	140 min (–77.4%)	9 / 10	2	50 min	400 u
Balance — delay/cost trade-off	150 min (–75.8%)	9 / 10	1	60 min	270 u
Zero transfer — minimal disruption	560 min (–9.7%)	7 / 10	0	150 min	300 u

Three lessons measured on S1:

- **Inaction also costs money:** 620 min of delay *and* 300 u of overtime. The Balance strategy beats it on both fronts (–75.8% delay, 270 u).
- **The value comes from constrained multi-line reallocation:** the best no-transfer plan recovers only 9.7% of the delay. The combinatorial search — the AI — is what creates the gap.
- **There is no single optimum:** Speed gains 10 minutes over Balance but costs 48% more — a trade-off that belongs to the manager, not to the algorithm.

4.2 Business assumptions (to be validated with an industrial mentor, then in the pilot — none presented as measured fact)

- Manual replanning after an incident: on the order of one to several hours; CableTwin targets a few minutes from incident to validated plan.
- One hour of extrusion-line downtime has a significant full cost (lost production, late penalties, overtime) — order of magnitude to be calibrated per site during the pilot.
- Several significant unplanned stops per month and per workshop — frequency to be measured in the pilot.

4.3 Pilot success thresholds (go/no-go targets agreed with the partner *before* measurement — validation targets, not achieved claims)

- **At least 30%** reduction in the time needed to assess an incident and compare recovery plans (operational target: < 10 minutes from incident to validated plan).
- **At least 20%** reduction in cumulative delay when replaying the site's historical incidents against the decision actually taken.
- **Zero hard-constraint violations** in plans presented as feasible.
- **At least 70%** of recommendations judged usable by the planner without a major rewrite.
- **100%** of approved decisions traceable to their inputs, assumptions and approver.

5. Feasibility

Proof by doing: the core of the product already works. The V0 engine — plant model, constraints, plan enumeration, multi-objective scoring, selection of the three strategies — was coded, run and versioned

today, on day 1. The figures in §4 come straight out of it, reproducible in one command (Appendix A). The remaining technical risk for Phase 2 is the interface, not the algorithmics.

Design choice	Why it makes the project feasible and demonstrable
Standard Python, zero dependencies for the engine; light web interface	Nothing fragile to install; the demo runs on an isolated machine.
Offline and deterministic	The live demo (Phase 3) depends neither on the network nor on randomness; same inputs → same plans.
Bounded but complete scenario (3 lines, incident, 3 strategies)	The full decision journey is demonstrable end-to-end in 90 seconds.
One-click reset	Replayable at will in front of the jury, with no residual state.
Read-only (CSV, then ERP/MES connectors)	The path to the pilot requires no writing into the customer's systems.

Execution credibility. The most tangible proof is already in this document: the engine that produces every figure in §4 was designed, coded, run and versioned within the first hours of day 1 — a deterministic, reproducible pipeline, not a fragile hackathon script. That pace rests on the solo founder's profile: 17 years of senior software engineering, combined with an ongoing academic trajectory at SUPCOM — a Master's in Machine Learning Engineering and a PhD in Artificial Intelligence — at the exact intersection of AI planning research and maintainable industrial software. Together, these sharply reduce the execution risk of both the Phase 2 prototype and the commercial pilot trajectory.

Execution plan over the three challenge days:

- **Day 1 (today)** — consolidated idea, runnable V0 engine, reproducible S1 figures. *Done.*
- **Day 2, July 18 (Build & Qualify)** — end-to-end prototype (the §2 journey in an interface, with test credentials), technical data room, 2-minute video, GitHub repository, brand kit, final deck; pre-selection on deliverables + video.
- **Day 3, July 19 (Polish & Pitch Lab)** — demo hardening, pitch training.
- **Grand Final, July 21** — 7-minute pitch + 4-minute Q&A, replayable live demo before the jury, executives and investors.

Risks and responsible deployment

Risk	Mitigation
Live-demo hazard	Offline, deterministic, one-click-reset demo, with a backup video.
Synthetic constraints differ from reality	Label them as such, validate them with a planner (industrial mentor), then replace them in a pilot.
At real scale, a meta-heuristic offers no global-optimality guarantee	Show several options, state the property, benchmark later against exact CP-SAT-class solvers (V0 is a complete search, hence exact on S1).
Repair duration is uncertain	Keep it visible and recompute the strategies when the estimate changes.
Sensitive production data	Minimal read-only files, local deployment, role-based access.
A real company is mistakenly implied	Fictional factory; manufacturers are cited as public sector references only (see References).
Credibility of figures	Strict separation of simulation / assumptions / pilot thresholds (\$4); figures reproducible in one command.

6. Originality

What CableTwin is not — not a dashboard that describes the past, not a 3D factory animation, not a generic LLM assistant, not an automation that silently changes production. What it is:

- **"Incident-first"** — planning tools optimize steady state; CableTwin is designed for the moment of crisis, where the value is at stake (and where today only the phone and the spreadsheet exist).
- **Comparability as the product** — the unit of value is not "an optimized schedule" but *three comparable decisions*: a decision-support tool built for an accountable manager, not an optimizer that imposes its solution.
- **Native governance** — mandatory human validation and an audit log from V0 onwards: rare in AI prototypes, decisive for industrial trust (and aligned with the spirit of AI regulations on operationally impactful systems).
- **Frugal, explainable AI** — constrained search + multi-objective scoring: no GPU, no external API, no randomness; every suggestion can be justified line by line.
- **Built "brownfield SME"** — one trade (cable), CSV exports, read-only: the product meets the factory as it is, not as Industry 4.0 dreams it.

7. Target audience

Role	Who	What CableTwin changes for them
Users	Production manager, planner, shop-floor supervisor	Decide in minutes, with costed alternatives, without rebuilding the schedule by hand.
Buyers	Industrial director, plant director, operations leadership	Fewer customer delays and less unplanned overtime; consistent, auditable decisions across teams.
Indirect beneficiaries	Sales/customer service, end customers	Delays known early, keepable promises, proactive communication.

Entry segment: cable and wire-harness factories with 3 to 20 lines and no APS — a dense fabric in Tunisia as well as in Europe and the Maghreb. The APII's overview of Tunisian industry (June 2026) reports **4,604 industrial enterprises** with at least 10 employees, of which 2,066 are totally exporting; the electrical, electronic and electrical-appliance sector counts **340 enterprises, 240 of them totally exporting** [2]. Internationally active cable groups such as Chakira Cables or COFICAB [3][4] illustrate the sector's density — cited here as public sector references only, with no affiliation; these figures describe the industrial base, not a guaranteed market for CableTwin.

The trade specialization (product families, line compatibilities, extrusion constraints) is an execution advantage: the V0 data model already speaks the shop floor's language. **Later extension:** any multi-line discrete manufacturing with dated orders (plastics, packaging, technical textiles) — after the value is proven on cable.

8. Competitive differentiation

Alternative	What it does	Its limit at incident time
Status quo — spreadsheet + phone	One handcrafted plan	Slow, blind to alternatives, untraced, dependent on one person.
ERP / MES (SAP, Odoo, industry MES)	Record orders and production	Systems of record: they observe the delay, they do not propose alternative plans.
Heavy APS (Siemens Opcenter, DELMIA Quintiq, Asprova)	Fine steady-state optimization	Expensive, months-long integration, dedicated expertise, black box — out of SME reach, and built for planning, not emergencies.
Generic AI copilots (LLMs over the data)	Plausible natural-language answers	No guarantee of constraint satisfaction, non-deterministic, hard to audit.
High-fidelity digital twins (physical/3D simulation)	Rich equipment and process simulation	Long, costly, physics-centred projects — not replanning decisions; CableTwin starts with the smallest operational twin that can improve one decision.
CableTwin	The empty slot between the ERP and the APS: the emergency decision. Three comparable plans in minutes, constraints guaranteed by construction, explainable and deterministic, human validation + audit, read-only, deployable at SME scale.	

Defensibility: the moat is not a secret algorithm; it is (1) the cable-trade business model (constraints, families, compatibilities) refined site after site, (2) the trust created by explainability and audit, (3) over time, the replayed incident history — a data asset neither the ERP nor the spreadsheet builds.

9. Business model and pilot trajectory (working hypotheses)

Envisioned model — presented as a hypothesis, with no unproven price or ROI claims: a paid pilot on one workshop or line, then a B2B subscription per site, sized on the number of lines, users and ERP/MES connectors; complementary services for integration and constraint configuration.

The pilot product the buyer actually purchases (hypothesis — no price, revenue or ROI claimed)

The economic buyer (plant / operations director) is not asked to replace ERP, MES or APS. They purchase a **time-boxed, read-only decision pilot** on one workshop:

- **Inputs:** historical incidents as CSV/export, under a minimal data contract — orders {id, family, duration per eligible line, due, priority}; lines {id, compatible families, busy windows}; incident {timestamp, stopped line, estimated repair duration}; and the decision actually taken that day (comparison baseline). This contract already matches the V0 engine's data model (Appendix A): the replay is pluggable with no custom development.
- **Method:** for each incident, freeze the same information the planner had at the time; generate the three recovery profiles; score CableTwin's options against the recorded decision on cumulative delay, late priority orders, overtime minutes and line changes.
- **Governance:** zero write-back to production systems; every recommendation requires human approval; full audit trail.
- **Duration:** 4–6 weeks; success thresholds agreed *in writing before measurement* (§4.3).
- **Commercial gate:** a per-site subscription (+ connector/configuration services) is discussed only if the thresholds are met; if not, the pilot stops — no forced integration.

This is how CableTwin becomes commercially survivable after the hackathon: paid learning on the customer's own incidents, not a promise of autonomous factory control.

Six-step entry strategy:

1. **Read-only** pilot at a first site (no writing into existing systems);
2. CSV import, then a read-only ERP/MES connector;
3. **Replay of the site's historical incidents** inside CableTwin;
4. Comparison of proposed plans with the decisions actually taken;
5. Measurement of replanning time and avoided delays (KPIs of §4.3);
6. Extension (more lines, more sites) only after demonstrated value.

This path asks the customer for no risky integration and no change to their systems: the first value is proven on their own historical data, within weeks.

10. Alignment with the official evaluation grid

Official criterion	Weight	CableTwin's answer
Business-process relevance and measured impact	25%	A real, critical decision (the line stop), a precise persona, a quantified engine proof with its scale stated (620 → 140/150 min on S1), and industrial impact defined only by the pilot's go/no-go thresholds (§1, §4).
AI integrated at the core of the product	20%	Constraint-based plan generation + multi-objective scoring <i>is</i> the product; includes the demonstrated case where the engine beats the human plan (§3).
Functional prototype	20%	V0 engine already runnable on day 1; complete end-to-end journey, offline, deterministic and resettable, delivered in Phase 2 (§5).
Economic viability	20%	User and buyer identified, dense entry segment, six-step read-only pilot, B2B subscription model as hypothesis (§7, §9).
Pitch and clarity	15%	A universal analogy (Waze), a one-sentence central message, a 90-second demo, figures with three levels of honesty (§2, §4).

11. Why this idea should advance

Against the four qualities expected of a Phase 1 idea, CableTwin is:

- **Bold** — it turns a lightweight digital twin into an operational decision product, where the market offers only systems of record or heavyweight optimizers;
- **Useful** — it addresses the production manager's real, stressful workflow, with human validation and traceability built in;
- **Measurable** — every option is compared on service, delay, overtime, transfers and extra cost; the figures are computed, reproducible and honestly labelled;
- **Achievable** — the engine that produces every figure in this document is already coded, versioned and runnable offline on day 1; the end-to-end prototype follows in Phase 2.

What we ask of the jury: select CableTwin for Phase 2, where the full journey — from the L2 stop to the audit log — will be demonstrated end-to-end, artifacts included.

References

- [1] Official site and page: automate-or-die.netlify.app · communitydays.org/event/2026-07-17/ai-and-automation-hackathon-automate-or-die; official phase and theme instructions received by the team leader's email (July 17, 2026).
- [2] APII, Overview of Tunisian Industry (source June 2026) — tunisieindustrie.nat.tn/en/tissu.asp (figures verified on July 17, 2026).
- [3] Chakira Cables, public overview — chakira-cables.com.
- [4] COFICAB Group, public profile — coficab.com.

Disclosure: CableTwin is not affiliated with any cable manufacturer (Chakira Cables, COFICAB, Elloumi Group or any other). The scenario's factory is fictional; all prototype data and all demonstrated results are synthetic.

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Appendix A — Synthetic scenario S1 and reproducibility

Synthetic workshop with 3 lines and **10 orders**: 7 committed orders (started or frozen, not replannable — they materialize the lines' occupancy) and 3 affected orders, planned on L2 and not yet started at breakdown time. Times in minutes from 06:00; shift end at 720 (18:00); overtime capped at 180 min/line (2 u/min); line transfer: 30 min setup, 150 u. **Incident**: L2 breakdown at t=120 (08:00), repair completed at t=600 (16:00). In the nominal (no-incident) plan, all 10 orders are on time, zero delay.

Line	Role	Busy until	Compatible families
L1	LV extrusion — line 1	t=290	F1
L2	LV extrusion — line 2 (versatile, down)	repair t=600	F1, F2, F3
L3	MV extrusion — line 3	t=620	F2

Committed orders (not replannable)	Line	Slot	Due
CMD-101 · CMD-102	L1	0–150 · 150–290	210 · 350
CMD-105 · CMD-106 (completed before the breakdown)	L2	0–60 · 60–120	120 · 180
CMD-103 · CMD-104 · CMD-110	L3	0–200 · 200–410 · 410–620	260 · 470 · 680

Affected order (planned on L2)	Family	Duration per line	Due
CMD-107 — LV cable 3×2.5 mm ² (9 km, auto OEM)	F1	L1: 90 · L2: 90	t=420
CMD-108 — MV cable 12/20 kV (4 km, energy)	F2	L2: 120 · L3: 120	t=630
CMD-109 — shielded instrumentation cable (2 km)	F3	L2: 60	t=700

The engine performs a complete search of the feasible space (compatible assignments × sequencings × overtime bounds): **11 feasible plans**, all scored; the three strategies are selected by objective profiles (Speed: minimal delay; Balance: normalized delay/cost trade-off; Zero transfer: minimal disruption). Full schedules in `sim/results.json`.

Reproducing the figures (Python ≥ 3.10, no dependencies):

```
python sim/cabletwin_sim.py — readable summary · python sim/cabletwin_sim.py --json — full results
```

Transparency note: the hand-built "Balance" plan during design was worth 170 min; the exhaustive search found 150 min. We publish the computed figure, not the hoped-for one — the rule we impose on every figure in this project.

Appendix B — Compliance with Phase 1 requirements

Requirement	Addressed in
The problem	\$1
The solution	\$2
The innovation	\$3
The quantified impact (simulation / assumptions / pilot KPIs separated)	\$4 (4.1 / 4.2 / 4.3)
The feasibility	\$5
The originality	\$6
The target audience	\$7
The competitive differentiation	\$8